Threat Analysis and OSINT Report

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Introduction

This report examines the cyber threat landscape of the UK Higher Education (HE) sector and assesses the security of a specific institution using Open-Source Intelligence (OSINT). The University of Cambridge has been selected for this investigation because of its importance as a major research university. With a group income of over £2.66 billion and a population of more than 25,000 students **[1]**, the University holds large amounts of valuable Intellectual Property (IP) and Personally Identifiable Information (PII). This scale makes it a huge target for both financially motivated cybercriminals and Advanced Persistent Threats (APTs) looking to steal research.

The primary aim of this report is to assess the current sector-wide threats (such as the 91% attack prevalence reported in the 2025 UK Government Cyber Security Breaches Survey **[2]**) and contextualize these risks specifically for the University of Cambridge. Through a passive OSINT investigation, we have mapped your digital footprint to identify key security vulnerabilities without using any active scanning methods against your infrastructure.

Cyber Threat Intelligence

The HE sector manages high-value IP and PII, attracting both state-sponsored espionage and financial extortion. The 2025 Cyber Security Breaches Survey confirms this severity: 91% of HE institutions identified attacks in the last year. Despite a marginal decline from 97% in 2024 **[5]**, the intensity remains critical, with 30% of institutions facing weekly incursions and 40% suffering negative outcomes, such as data loss or service disruption. **[2]**

This high frequency is driven by Advanced Persistent Threats (APTs) like Silent Librarian (Cobalt Dickens). Affiliated with the Iranian-based Mabna Institute, this actor has systematically targeted university research since 2013 **[4]**. Their reliance on credential theft via spoofed login pages correlates with survey findings identifying phishing as the dominant vector, affecting 97% of institutions **[2]**.

Beyond phishing, the attack surface is diversifying. Supply chain attacks have risen significantly to 69%, forcing institutions to audit third-party software rigorously. Other critical vectors include impersonation (68%) and Denial of Service (36%). While ransomware remains lower volume (15%), its impact is often catastrophic. However, defences are hardening; 94% of institutions now conduct phishing simulations, compared to just 18% of businesses **[2]**. These threats are global, with similar campaigns compromising US Ivy League institutions like Princeton and Harvard **[3]**. These

precedents underscore the specific risk to UK equivalents like Cambridge, confirming that research-intensive universities are prime targets for data exfiltration.

Furthermore, universities are harder to defend than standard businesses because they are designed to be open and accessible. Unlike an organisation that owns and monitors each device, universities must allow students and staff to bring in and use their own devices, making it more difficult to secure **[7]**. Additionally, institutions like Cambridge produce technology that can be used for both civilian and military purposes, making the university a target for foreign governments **[8]**. Instead of developing this technology themselves, these actors try to steal data directly from university servers to bypass international trade laws **[9]**.

OSINT Analysis

This investigation strictly employed passive reconnaissance techniques to map your digital footprint. The objective was to identify public exposures that a Threat Actor exploit, without directly engaging or scanning your infrastructure. This analysis followed the Cyber Kill Chain reconnaissance phase: gathering intelligence, mapping the network, and identifying potential entry points.

Using Google Advanced Search operators, we aimed to uncover sensitive directories (site:cam.ac.uk). While standard filetype queries returned vast amounts of benign academic data, refining the search to administrative keywords (filetype:pdf "admin" "guide") successfully exposed your internal "Equipment Sharing Database User Guide" **[A].**

While the document itself is not classified, it provides a "blueprint" for an attacker. It explicitly details user roles, internal terminology, and confirms the University's reliance on "Raven" for Single Sign-On (SSO) **[6]**. For a cybercriminal, this is valuable context; knowing exactly how the internal systems are named and structured allows them to craft highly authentic and convincing spear-phishing emails targeting your staff.

To understand your underlying web technologies, Shodan.io was used to profile your servers. The analysis revealed a significant migration to third-party cloud providers, with many services hosted on Google Cloud Platform (GCP) and Amazon Web Services (AWS).

Crucially, the query identified multiple subdomains labelled webapp.devel. These potential non-production environments are a frequent source of risk, as they historically lack the rigorous security patching applied to live sites, production sites. Furthermore, while cloud infrastructure is highly resilient, rapid deployments and migrations frequently lead to misconfigured storage assets. For a threat actor conducting passive reconnaissance, identifying misconfigured, publicly accessible AWS S3 buckets or GCP Storage blobs tied to these development subdomains represents a critical vulnerability **[B]**.

Furthermore, passive header analysis detected Apache HTTP servers (v2.4.52/2.4.58) issuing mod_auth_openidc cookies **[C].** This confirms your use of the OpenID Connect protocol for your central login system. While efficient, this creates a high-stakes security model: because one Identity Provider handles logins for numerous university services, the compromise of this single system would yield catastrophic, university-wide access for an attacker.

A Censys query returned over 58,000 results, indicating a massive digital footprint **[D]**. Automated tools (theHarvester and DNSdumpster) were used to map out the subdomains that sit beneath your main university website. Two specific areas of risk were identified:

1. **Administrative Interfaces:** Subdomains like hive.em.admin.cam.ac.uk suggest backend staff interfaces. If exposed publicly, these are high-value targets for credential theft. **[E]**
2. **Development Environments:** The network map highlighted subdomains containing "dev" and "api," such as api-dev-amicus.admin.cam.ac.uk **[H, I]**. This is a critical finding. "Dev" (development) environments are test sites often used by IT staff before a system goes live. If an attacker can find a vulnerability here, they may be able to use it as a "backdoor" into the wider network.

A "Whois" query confirmed you successfully utilize a privacy service to redact administrator contact details **[F]**. This is a strong positive security control, preventing attackers from easily scraping specific IT staff names for social engineering campaigns. The records also confirmed your connection to the JANET network., linking it to the wider UK academic infrastructure.

Finally, nslookup analysis confirmed a "dual-stack" configuration, making you accessible via both legacy (IPv4) and modern (IPv6) addresses **[G]**. While necessary for modern connectivity, this effectively doubles your attack surface. If your security teams have configured robust firewalls for IPv4 but lack equal maturity and visibility over IPv6 ranges, these newer addresses could serve as an obscured entry point for the attacks described in the threat intelligence sector analysis.

Discussion

The correlation between sector-wide threat intelligence and the specific OSINT findings reveals a critical vulnerability in your security posture. Your reliance on the "Raven" SSO system creates a single point of failure. If a threat actor successfully spear-phishes a single administrator credential, they bypass your perimeter entirely.

Furthermore, your ongoing migration to AWS and GCP introduces the severe risk of data exposure through misconfigured cloud storage. If an S3 bucket is left public during the development of your webapp.devel subdomains, threat actors could passively scrape sensitive institutional data or internal credentials without ever needing to breach the Raven SSO perimeter.

While this technical analysis focused on infrastructure, a comprehensive threat actor would likely pivot to social media reconnaissance (e.g., LinkedIn) to identify specific Raven users. Targeting these individuals with phishing emails remains the most probable path for credential theft. Fortunately, your Whois privacy redaction successfully breaks the "Reconnaissance" phase of the Cyber Kill Chain by masking administrator details. This obfuscation forces attackers to rely on broader, less effective phishing campaigns rather than targeting specific high-privilege IT staff.

However, the infrastructure analysis highlights the persistent risk of your development subdomains. This affords an attacker two distinct avenues: brute-force the hardened Raven login, or attempt to slip through a less-secure development API. Current CTI data suggests financially motivated groups will likely exploit the API to deploy payloads (like ransomware), whereas state-sponsored actors will target the administrative system via Raven to establish long-term, persistent access.

Finally, your IPv4/IPv6 configuration creates a dangerous potential opening. When legacy security tools fail to monitor IPv6 traffic, they provide blind spots for Command-and-Control (C2) traffic to operate undetected. This lack of visibility is a primary reason why 30% of HE institutions now report successful weekly attacks; unmonitored edges allow automated scanning to easily transition into persistent security breaches.

Conclusion

This report confirms that the UK Higher Education sector, and the University of Cambridge specifically, faces a severe and evolving threat landscape dominated by credential theft (97%) and ransomware. This OSINT investigation reveals that your security posture is strong regarding administrative privacy, but fragile regarding your rapidly expanding technical attack surface.

Your reliance on the Raven SSO system creates a critical dependency: a single compromised credential could grant wide-ranging access to highly sensitive internal data. Furthermore, the discovery of development subdomains and API endpoints exposes a "Shadow IT" risk that inherently bypasses traditional perimeter defences.

To remediate these vulnerabilities, we recommend you immediately audit your cloud environments (AWS/GCP) and ensure that all non-production and development assets are subjected to the exact same rigorous security monitoring, access controls, and patching schedules as your live systems.

References

[1] University of Cambridge, "Reports and Financial Statements for the Year Ended 31 July 2025," *cam.ac.uk*, 2025. [Online]. Available: https://www.cam.ac.uk/system/files/university_of_cambridge_group_annual_reports_financial_statements_2024-25.pdf. [Accessed: Feb. 2026].

[2] Department for Science, Innovation and Technology, "Cyber security breaches survey 2025: education institutions findings," *GOV.UK*, 2025. [Online]. Available: https://www.gov.uk/government/statistics/cyber-security-breaches-survey-2025/cyber-security-breaches-survey-2025-education-institutions-findings. [Accessed: Feb. 2026].

[3] Nature, "Cyberattacks on US universities are on the rise," *Nature*, vol. 638, 2025. [Online]. Available: https://www.nature.com/articles/d41586-025-03484-9. [Accessed: Feb. 2026].

[4] MITRE ATT&CK, "Group G0122: Cobalt Dickens," *MITRE*, 2024. [Online]. Available: https://attack.mitre.org/groups/G0122/. [Accessed: Feb. 2026].

[5] Department for Science, Innovation and Technology, "Cyber security breaches survey 2024: education institutions annex," *GOV.UK*, 2024. [Online]. Available: https://www.gov.uk/government/statistics/cyber-security-breaches-survey-2024/cyber-security-breaches-survey-2024-education-institutions-annex. [Accessed: Feb. 2026].

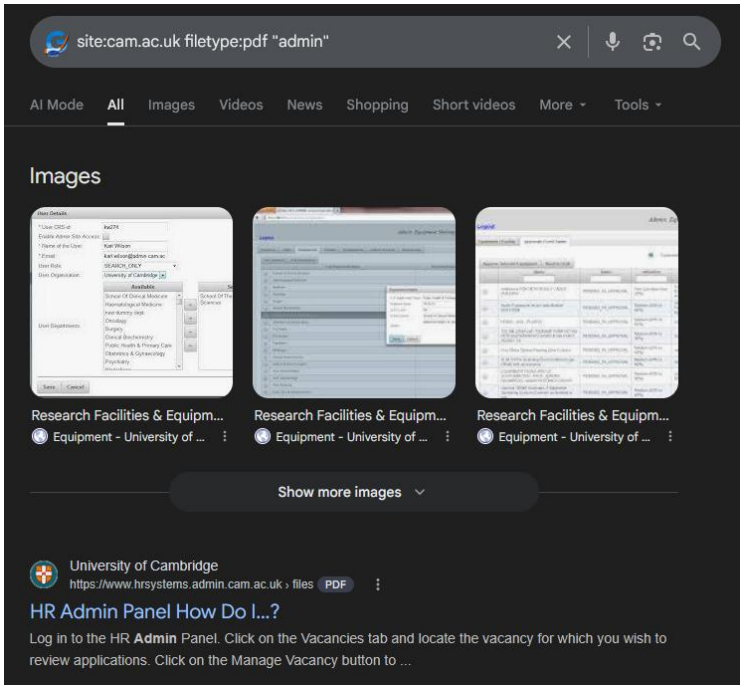
[6] University of Cambridge, "Equipment Sharing Database User Guide," *admin.cam.ac.uk*, n.d. [Online]. Available: https://www.equipment.admin.cam.ac.uk/esd-admin/raven/Equipment%20Sharing%20Database%20User%20Guide.pdf. [Accessed: Feb. 2026].

[7] Jisc, "Security Conference 2025: Why it's time to act on tackling cyber threats," *Jisc*, 2025. [Online]. Available: https://www.jisc.ac.uk/blog/security-conference-2025-why-its-time-to-act-on-tackling-cyber-threats. [Accessed: Feb. 2026].

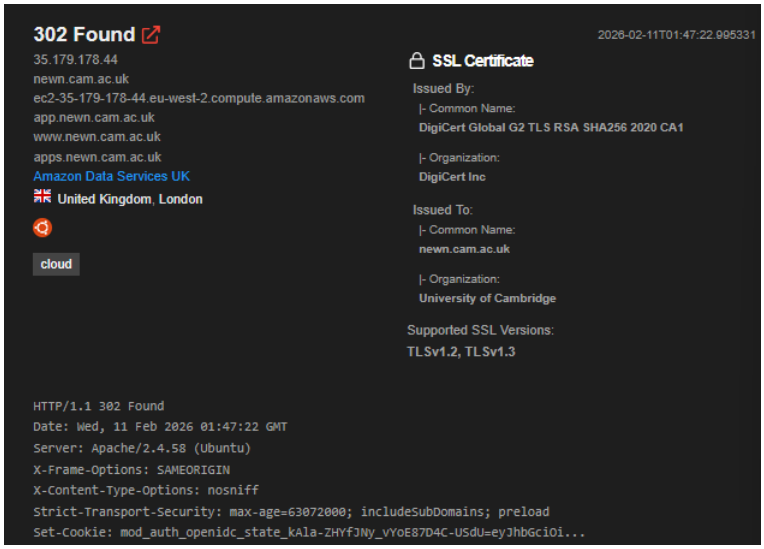
[8] National Protective Security Authority (NPSA), "State Threats in Academia: Secure your success," *NPSA*, 2025. [Online]. Available: https://www.npsa.gov.uk/national-security-act/state-threats/state-threats-academia. [Accessed: Feb. 2026].

[9] Royal United Services Institute (RUSI), "The Government Needs to Send a Wake-Up Call on Research Security," *RUSI*, 2025. [Online]. Available: https://my.rusi.org/resource/the-government-needs-to-send-a-wake-up-call-on-research-security.html. [Accessed: Feb. 2026].

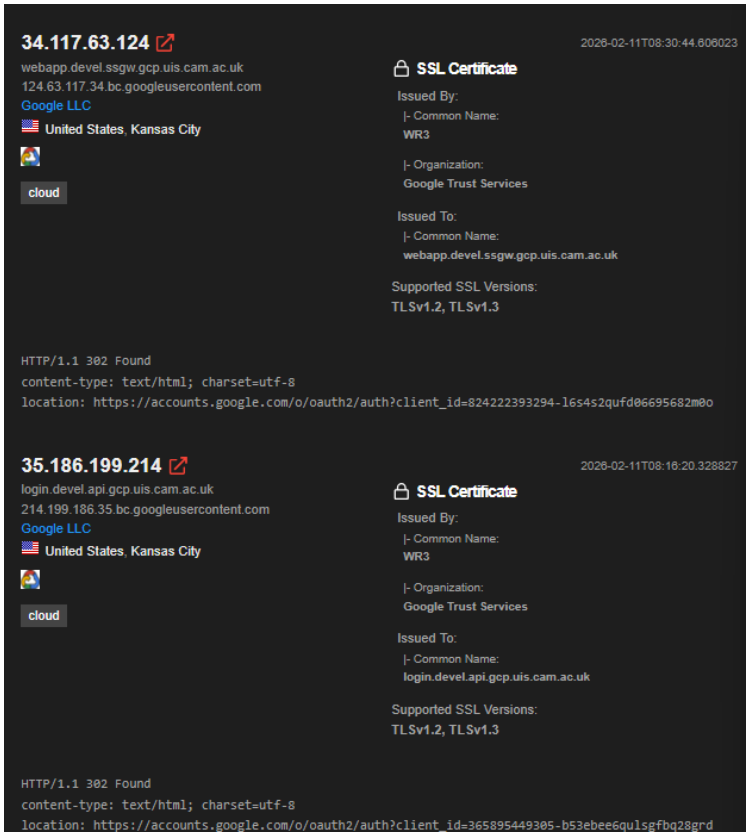
Appendix



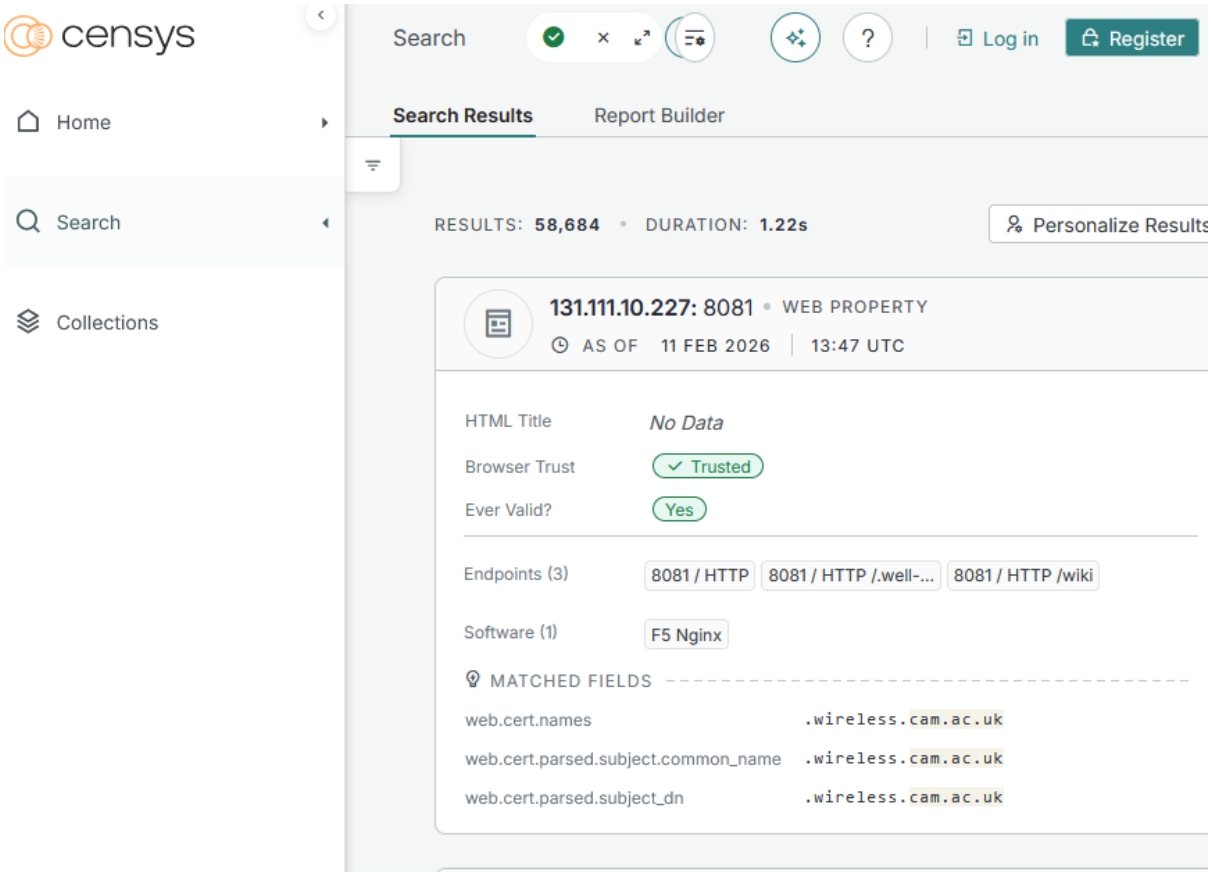
[Appendix A]



[Appendix B]



[Appendix C]



[Appendix D]

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*******************************************************************
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* | __|  _ \ / _ \  / /_/ / _` | '__\ \ / / _ \ __| __/ _ \ '__| *
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*  \__|_| |_|\___| v /_/ \__,_|_|    \_/ \___||___/\__\___|_|     *
*                                                                 *
* theHarvester 4.8.2                                              *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************

[*] Target: cam.ac.uk

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
        Searching 0 results.
[*] Searching Bing.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] Hosts found: 1
____________________
hive.em.admin.cam.ac.uk
```

[Appendix E]

```
Domain:
        cam.ac.uk

Registered For:
        The University of Cambridge

Domain Owner:
        University of Cambridge

Registered By:
        Jisc Services Limited

Servers:
        auth0.dns.cam.ac.uk     131.111.8.37
        auth0.dns.cam.ac.uk     2001:630:212:8::d:a0
        dns0.cl.cam.ac.uk       128.232.0.19
        dns0.cl.cam.ac.uk       2a05:b400:110::d:a0
        dns0.eng.cam.ac.uk      129.169.8.8
        ns1.mythic-beasts.com
        ns2.ic.ac.uk
        ns3.mythic-beasts.com
        auth1.dns.cam.ac.uk     131.111.12.37
        auth1.dns.cam.ac.uk     2001:630:212:12::d:a1

Registrant Contact:
        Cambridge Hostmaster

Registrant Address:
        Cambridge University Information Services
        Roger Needham Building
        7 JJ Thomson Avenue
        Cambridge
        Cambridgeshire
        CB3 0RB
        United Kingdom
        +44.1223748440 (Phone)
        hostmaster@cam.ac.uk

Renewal date:
        Thursday 13th Aug 2026

Entry updated:
        Tuesday 3rd February 2026

Entry created:
        Wednesday 17th September 2003
```

[Appendix F]

```
Server:         ::1
Address:        ::1#53

Non-authoritative answer:
Name:   cam.ac.uk
Address: 23.185.0.3
Name:   cam.ac.uk
Address: 2620:12a:8001::3
Name:   cam.ac.uk
Address: 2620:12a:8000::3
```

[Appendix G]

[Appendix H]

api-dev-amicus.admin.cam.ac.uk	131.111.150.129 amicus.admin.cam.ac.uk	ASN:786 131.111.0.0/16	JANET Jisc Services Limited, GB United Kingdom	https: unknown server cn: amicus.admin.cam.ac.uk	5
api-train-amicus.admin.cam.ac.uk	131.111.150.129 amicus.admin.cam.ac.uk	ASN:786 131.111.0.0/16	JANET Jisc Services Limited, GB United Kingdom	https: unknown server cn: amicus.admin.cam.ac.uk	5
api-uat-amicus.admin.cam.ac.uk	131.111.150.129 amicus.admin.cam.ac.uk	ASN:786 131.111.0.0/16	JANET Jisc Services Limited, GB United Kingdom	https: unknown server cn: amicus.admin.cam.ac.uk	5
api-uat-symplectic.admin.cam.ac.uk	131.111.150.140 api-uat-symplectic.adm	ASN:786 131.111.0.0/16	JANET Jisc Services Limited, GB United Kingdom		1

[Appendix I]

Student Declaration of AI Tool use in this Assessment

Please indicate your level of usage of generative AI for this assessment - please tick the appropriate category(s).

If the "Assisted Work" or "Partnered Work" category is selected, please expand on the usage and in which elements of the assignment the usage refers to.

Solo Work	**S1 - Generative AI tools have not been used for this assessment.**	☐
Assisted Work	**A1 – Idea Generation and Problem Exploration** **Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.**	☐
	A2 - Planning & Structuring Projects **AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.**	☐
	A3 – Code Architecture **AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.**	☐
	A4 – Research Assistance	☐

	Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions. The interpretation and integration of research into the assignment remain the student's responsibility.	
	A5 - Language Refinement **Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.**	☒
	A6 – Code Review **AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.**	☐
	A7 - Code Generation for Learning Purposes **Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.**	☐
	A8 - Technical Guidance & Debugging Support **AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.**	☒
	A9 - Testing and Validation Support **AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.**	☐
	A10 - Data Analysis and Visualization Guidance **AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.**	☐
	A11 - Other uses not listed above **Please specify:**	☐

Partnered Work	**P1 - Generative AI tool usage has been used integrally for this assessment** **Students can adopt approaches that are compliant with instructions in the assessment brief.** **Please Specify:**	☐

Please provide details of AI usage and which elements of the coursework this relates to: **A5: Used for spelling and grammar checks across the report. A8: Used to clarify technical concepts after use of tools.**

I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.	☒
I confirm that all details provide above are an accurate description of how AI was used for this assessment.	☒